

Hello everybody.

Thank you again for your advice about my interview with Ed Scheidt.

I talked to him last week. My impressions are

1) He is a very nice person! Very gracious.

2) His answers did not seem overly helpful. They were kind of dense and discursive. Part of the problem is that I'm not enough of an expert in cryptography. And part of the problem is that he often speaks in abstract and vague language. Specifics were few and far between.

As for the big question about whether he'd solve K4, he said this:

He will not try to solve K4 because he likes not knowing the answer when he is talking to journalists or solvers. It means he doesn't have to lie. He said that his experience in the CIA has taught him ignorance is sometimes the best course.

I am including the transcript. Again, I don't think there's a huge amount to be gleaned from it. But maybe those who are sharper than me can find a sliver.

If folks have further questions, I am happy to try to get him back on the phone. He seemed open to talking again. Just let me know.

Thanks!

AJ:

Would you mind taking it from the start, how you got asked to be involved?

Ed Scheidt:

Well as you may have read, I was an employee at one time of the CIA, and I was also reaching my retirement age and others knew that, so I was asked by the senior people if it was possible to create something that would be more on the commercial side just for defining things, because I worked in the world where you had terminology assets and all this other stuff. Of course the government is not interested in giving out how they talk to different people and so forth. So I said sure why not, and then the next step was for me an interesting step, and the step was, what would you envision, how hard ... because cryptography can come in different gradations of difficulty. And so I said how hard? And they said well don't make it too hard, T-O-O. And I said, well that's a subjective thing but that's all right too. So I left and came back and said well I have some suggestions and see what you think. In my travels and experiences and so forth I've come across different languages and things like this, if you are going to go a commercial type of thing and you want to give some opportunity for people to ... we'll use the term break the system or be able to hack inside into it, the choice of a language becomes important.

Ed Scheidt:

And so I started with a language that was really only used in the world for about 100 years, that was defined a dead language today-

AJ:

Would you mind backing up for one second, you said a choice of a language...

Ed Scheidt:

Yeah the choice was do I ... well I presented some choices and my choice was a language that was only used in the world for maybe 100 years or so. Then it was the community that used the language disappeared. I don't know the rest of the story but ... so that was one, and then I suggested well, I enjoy opportunities to teach cryptography and stuff like that, perhaps I could portray a educational aspect of it, and use some traditional cryptography that dates back to say 1700s I suppose, and up to modern usage. But this is a paper and pencil design, that what you see today which is the digital design, so the digital designs have inherent fundamentals, but don't exhibit the fundamentals like I was going to do and such. So I said that, and then I would suggest we used the English language as a reference. Now why becomes important in the discussion of, if you're going to give someone the opportunity to do analysis of your work, you wind up with what you may remember or hearing a reason that the Navajos were used during World War II is the United States didn't believe anyone had cataloged that particular language.

Ed Scheidt:

And if you don't catalog it that means before you get to the crypto part which is the math part, you still have to deal with the language. That was ... what was it written in? And to break Kryptos, the various aspects that had taken place, they used that fundamental principle. Now if we have more ease in the English language than any other letter. We have few Zs in the English language, therefore you wouldn't start your analysis based on Zs. So that's the principle, and from there you have an opportunity and the statistics can work with you if you take that approach, which people who are in the business of crypto-analysts, crypto-analytics definitely know this type of a principle.

Ed Scheidt:

So after discussion and so forth the agency people said well the intent is to have it so that the design could be broken in time. At that time it was looking at three to five years, that was a matter of genius as they say, and stuff like that. But I said well there's some tricks I can do as long as I can get permission to do it. I recognize where I'm sitting.

Ed Scheidt:

So then I retired, that left it so that they didn't have to pay me [inaudible 00:13:25] if I wasn't retired then I'd get my employee's money. And so I said okay I'm willing to do this free. So I didn't hardly get any money from it. So, that's the genesis of it. And that led into the, can you do an educational type of presentation while you are creating a secret? And one of your questions that you needed to ponder is, why does everybody have an interest in this 30 years later?

AJ:

Great question.

Ed Scheidt:

Yep, and so a lot of this genesis that I'm mentioning to you is just a stage for a while, because part of what has transparent is transparent is you have a secret. Surely the secret is discoverable and I want to be the discovery, because as you are writing a book, and you're looking to do it in a broad sense, at one time it was 3,000 people working on this. Not with me. So there's clubs if you talk to Elonka and so forth you know that there's various groups, clubs that have met. I have had one that could do it, I did for a bunch of years, once a year in October invited people who had interest in Kryptos, we would go to a restaurant for a while and then a couple of more at my house, because I can accommodate say 20 people, something like that. And it was not in a formal supper but sandwich and stuff like that. But it was very fascinating because you're in one sense an adversarial world because everybody wants to know answer and ... or if there's more than one answer. Why? Because you can get very rich on this particular one.

Ed Scheidt:

At one time, and I do not keep up with this at all, but I was told that this particular puzzle has a genre in puzzles of the world, so it says if you broke that, ergo which drives the Elonka of the world, it's part of our credence too.

AJ:

It's always listed in the lists of great unsolved puzzles, it's always up there with the Voynich manuscript.

Ed Scheidt:

Well you also have other facets. One is, and this was part of the discussion before I was retiring. Are you going to keep a secret because of where the puzzle is located? It's in a courtyard of the CIA. The average person is not going to get in to see it. So it means are you willing agency to publish it, to illustrate it, to what extent you're going to want to talk about it? Because they use it also, or as I understand it, like a public relations. At one time I wrote the ... they used to have on the ... for recruitment, some CIA puzzles, totally separate of Kryptos, that was the person who put it together, for a presentation on the internet, and it was [inaudible 00:17:42] or whatever. Again it was to attract the younger folks who maybe have math backgrounds, or maybe are interested in being in analysis and stuff like that. So it's been used for the own internal way of saying, we have capabilities and here's an example of where it used as such.

AJ:

Now when I talked to Jim he mentioned that you two would meet at a restaurant near the headquarters and talk in a low hushed voices.

Ed Scheidt:

You know when we discovered, now you're moving into the actual production, if you want to call it the phase or work, you're going to put things together and so forth. Originally Jim thought that he could do the work and I could visit, he has a workspace in Maryland that he had used and [inaudible 00:19:11] media like yourself learned of that. Even though the quest is to keep things as secret, there are no real secrets in this world, and if you're a citizen. So I suggested to Jim why don't we do some techniques that could be used for the clandestine side of it, because it ties to the subject matter. So he agreed and that's where you lined up with restaurants, visited my house on occasion. Not that it's a clandestine site but it's not advertised as such, where his commercial site was advertised. So

that worked out well, the concern was for audio, mostly for audio as opposed to seeing you at a place or something like that. And as you know the media has audio capabilities and pick up voices from a distance. So you could be in a facility and they could still hear you, or what's going on.

Ed Scheidt:

The obvious is, well not necessarily obvious but the goal was to find out what was the secret, how did you approach the subject of the content part? Which we have a process, we have all the inner changes that build to that, and now we move into if you are going to portray something in the CIA, [inaudible 00:21:17] do, what are the specifics that have ... and at the same time you have the gunmen who would like to retain as much confidentiality up to the point of when they announce it and show you and stuff like that. So this went on for several months. And you're limited on what you can do, it's just like I'm being interviewed by ABC News many many years ago, 20 years ago or so, and they told me, well Ed, this is going to be an experience for you because it's in real time. Which means whatever you say, and at that time 25 years ago they didn't have the delay capability that you get today..

Ed Scheidt:

I said, and then ABC visited me at my house as sort of a formality, and said okay we're going to ask you these questions and lead me on in the interview to what you want, while they had certain things they wanted. And I answered them and stuff like that, and the punchline was, I get all wired up, I'm ready to start the interview in the sense of this is live on the ABC set in DC, so it's like a live set. Then I hear New York because it's New York feed, which is where the journalist was at the time. Well he introduced himself and stuff like that, and then in the background came... say, Ed everything we talk about this weekend forget because I'm not going to ask you these questions, I'm going to ask you what you would like to hear. Now think about that, you're live there, 20 million people around the world, you know ABC is big, and it's like the New York Times, a lot of people read it, and a lot of people hear. And I said, well okay this is an interesting experience.

Ed Scheidt:

So went through that, the good news is, people after it, they said well Ed you had your three minutes.

AJ:

So I think Jim had said, just like me, he was no expert in cryptography, and that you gave him a crash course. So what were some of the things you taught him.

Ed Scheidt:

He learned enough because I did the education as part of his result type of thing, in other words, we're going to come with an answer, let me show you how the answer. And then while we are doing that, let's talk about why did we choose this way of doing it, that's the education part. In other words I could sit here and say well, we're going to do substitution in the genre of cryptography, and you look at me and say, okay. That 25 cents gets me a cup of coffee. And that was ... it has no meaning. Where if I could say we're going to make a puzzle that you would like to have, and while we're making that puzzle let's introduce cryptography into the puzzle and then I will give you some reasons of why this was chosen to do it that way. And so that's some of the education side at Jim's level. Now in the overall project I looked at the way ... has taken place in modern cryptography, not digital now, but paper and pencil. Modern cryptography, what are the bases that make it up?

Ed Scheidt:

And there you are looking at mathematically how can you make something that will create a secret? Well that's the simple or broad way of addressing this. And then from there on, I said well let's use examples as well as create the puzzle. In other words, if the puzzle is only to be out there three to five years, then we can deal with that in the context of known, understand academic level of cryptography. And there you wind up with, in the genre substitution is the name that flies, you're going to substitute something for something else, since we're using the English language. You're going to substitute one letter for another letter, and what makes you substitute is part of the

math of cryptography. So we began with that, substitutions if you want to read a little bit about this, you'll find that that dates back a lot of time, in other words you take the alphabet and you just move it down one character or two characters and then you start combining the functions. And then you wind up with processes or process of encryption that is acceptable in the paper and pencil which was done in the 17th century.

Ed Scheidt:

And well understood again a mathematical problem that's been used for puzzles for many many years, well hundreds of years. So to take these various principles, in other words are we going to make something that AJ would understand, Jim has never been exposed to cryptography but yet I've got to teach him, but at the same time, you don't care. Two weeks after you've done this it's gone, it's not coming back, that's not his thing. So you've got to make something that's retainable, that you can write it on a piece of paper because that's what I did, and it can be justified by others that it's not stupid, and it has credence and so on. But it meets some criteria. So substitution was one I had to use, another math problem is transposition, transposition is again the name implies, you're transposing something. So instead of a direct correlation that you can visualize, transposition is a little harder to visualize. It's sort of like looking at a puzzle and you're defining the parameters of the puzzle in the sense of the square, and then now you have the square and you're going to transpose the characters or the letters that are in the square to something that's a secret.

Ed Scheidt:

So my secret is push into this square. So that was another step. And then the last step which has been good for 30 years, which I didn't know at the time it would be, but I masked the framework, in other words if you can change the language base then it becomes in my favor and not your favor of trying to break it. It becomes more of a challenge now, when it was used as the mask it was current, 2020 secret.

AJ:

As I mentioned, I am a beginner in cryptography so what does that mean to change the language base? And can you tell me more about masking?

Ed Scheidt:

Well mask ... mask is that have used the English to write not a problem. In other words here's what I want to keep secret, I write it in English. If I use that directly in the puzzle I have against me the number of Es, the number of Ys and number Xs and so forth [I wrote. Now if I can mask that, change that characteristic, then you as a breaker of the system whatever it is, different views of what you're looking at. You are either the maker of the system, like Sanborn, or you're the breaker of the system, which is the majority of the community of the world who would like to know what does that puzzle say? And is a puzzle inside of a puzzle? Because it can be that too. And so my son works for NBC and he talks to me about it, because I'll say that I had some talks or an interview of something like that. And he's heard the subject for 30 years but he's not the math of the family. That's just how we are, but he enjoys listening to the stories that have come out of this thing.

AJ:

Oh yeah well they wanted me to come and see it, but because of COVID that's delayed, so hopefully I'll go get to see it soon.

Ed Scheidt:

Either get yourself somewhat familiar because it's a sculpture, and looking at a sculpture by itself without a context it's not as much fun as ... if you look at this way, and there's other things that have been put out on the compound of the CIA in addition to the sculpture itself, and there's a been a lot of interest over the years, potential meanings of all these things.

AJ:

Yeah, can I ask again about the masking. How do you crack a masked code?

Ed Scheidt:

That's for me to know and for you to find out.

AJ:

Good one, but in general, not this one, but is there a key word or a key phrase or is it ...

Ed Scheidt:

No clues.

AJ:

No clues, all right. Well I'm not asking, I still am not sure what changing the language base means. Like you're not changing it to a different language like French or German, you're changing it to a ...

Ed Scheidt:

Something.

AJ:

What's that?

Ed Scheidt:

Something else.

AJ:

What's that?

Ed Scheidt:

Not to another language, it's to something else.

AJ:

Oh, so it could be to something else like something in world like the window pattern across that I'm looking at right now, that kind of thing?

Ed Scheidt:

It could be, but it's not.

AJ:

It's not the window pattern outside my window?

Ed Scheidt:

No, sorry.

AJ:

Well I thought I had cracked it.

Ed Scheidt:

That's the currency. You asked about what keeps people going. It's a puzzle, and if you figure out the mechanics of how you did it, then it gives a insight into the answer.

AJ:

Right, well I Google searched to see what are the hardest types of encryption, and people talk about the One Time Pad, so is that a different thing from masking, or is that related to masking?

Ed Scheidt:

It's not directly related but because you're trying to say, did you use a one time pad per se, that's not the intent. The analogy is reasonable, but reasonable in a sense of the result. How you did it to get that result is the question.

AJ:

Got it, and the other thing Jim mentioned was interesting, he said that you told him about codes that are woven into African blankets

Ed Scheidt:

Clothes that have part ties and other apparel, things like that. And I had a very interesting career, part of a career. Because how you portray the result is another story in itself. In other words, I've got a fancy code, but I have to be able to use it as such. And how I'm going to use it is another subject that enters into the puzzle as such. Because if I want to say I can put it in a blanket, okay, that means I carry the blanket with me. It means I can go across borders with that

blanket, and that can happen depending on your business opportunity, it has a meaning to you, or a meaning to a person. So it puts you into the use of these devices, and it lets you write books. Not necessarily in the context of what you're writing, but the war stories and all the other things. Because you're there, you're experiencing the result of doing this, if you are wanting to be able to have you talk in a different country to me how do you do that? Because if you are determined that you're talking to me, you die.

Ed Scheidt:

And therefore it's taken seriously, as opposed to casual.

AJ:

Yep, the stakes are high.

Ed Scheidt:

Yeah the stakes are high, can be, and they're stories. Thousands of stories along those lines.

AJ:

So have you thought of writing a book yourself?

Ed Scheidt:

I did but I had too many other things going on. I don't know if you ever heard of digital currency?

AJ:

Sure.

Ed Scheidt:

Well I'm heading up the security of digital currency for all of ISO, I don't know if you know what ISO is? International Standards.

AJ:

Right, I saw that. You are ahead of the curve on that one.

Ed Scheidt:

Cryptography is only a part of it.

AJ:

Right, but a huge part of it. One thing I wasn't clear on, did you know the answers to K1, K2 and K3? Or did you crack them or learn them when they were solved?

Ed Scheidt:

I learned them when it solved. How I work with people is, I firstly do not want to know and if the arrangement is I show you how to do it, you do it, now if you make a mistake in doing it, I'm not here to correct you. I can be I mean obviously I've been asked to review things. But in the case of Kryptos I tried to stay ... okay Jim you have the secret, you have the puzzle, and therefore when I get all these wonderful interviews, I don't have to lie to people.

AJ:

Ah, I was wondering why, aren't you curious to try to solve K4?

Ed Scheidt:

I've also learned in my experiences in years past with the government and all, it doesn't do you any good to be too curious. And in the case of ... I just shared the story with the real time, I'm trying to ... this sounds bad but I'm trying to vie my wits with a correspondent that could be much more smarter than I. In his head the ability to ask those tricky questions, and if you've got cameras there focusing on you as you're asking those tricky questions, you can determine he's lying, or he knows something that he's not sharing. Which is, in this case, is [inaudible 00:44:36] things. And then when I have like I said shared with you, you have a supper with a dozen, two dozen people and they're all looking at you, and they all have had a chance at home or otherwise to come up with questions, just like you focused on the mask. They ask six questions upon that. You're seeing that I have a high IQ too, so anyhow. And it's not wrong, it's just okay I'm not a person that likes to lie.

AJ:

So ignorance is the best defense. You can't be caught lying if you don't know it

Ed Scheidt:

If I don't know it, I can't make a mistake

AJ:

I asked a couple of people I know in the Kryptos community if they had any questions for you, so can I ask you just a couple of those? They wanted to know, was the book that you gave Jim, was that the Cryptanalysis by Helen Gaines?

Ed Scheidt:

No, I don't think it was, I don't remember that but I don't believe it was.

AJ:

Do you remember what book you gave him or what materials?

Ed Scheidt:

It was many years ago.

AJ:

How about this question, are you ready? This is one of my favorites. Can you arrange a meeting at the Tiger's Den at 7.30 am?

Ed Scheidt:

I don't think the Tiger's Den is around any more. But there was such a place, at least as I remember it years ago, that you could do that.

AJ:

It was a restaurant? I have no idea what it means, he just asked me to ask you.

Ed Scheidt:

Yeah, if I remember right, it's years ago, it was like restaurant. And it was just a gathering place, and it was a gathering place for people who had interests in Kryptos and wanted to talk about it. You have a transformation of being able to gather information, now you have world wide internets and stuff like that, so you can reach anybody almost anywhere so gathering the areas is one thing. The other is there's enough [inaudible 00:47:46] type of things that the internet is not thoroughly secret either, and because of that it's not as if you can get it in a social environment, like I said have a lunch or a drink or something like that. And then talk on the subject and as well as, what you didn't ask them is, what in their question does it include a secret that's related to them? Because it's a strange world that we live in, in other words Kryptos has secret, does AJ have a secret? That he would like to keep relating to Kryptos? Because the carrot is there, is if you could gain for instance [inaudible 00:48:47] your discussion or [inaudible 00:48:49] today, if you could gain how the fourth part what makes it up, or what's the direction and stuff like that. That's worth money to you, and it's worth fame to you, [inaudible 00:49:08] you may not have recognized it.

Ed Scheidt:

Because it is a hard puzzle, and today. That wasn't it's intention 30 years ago. But because when you are making things in the world of cryptography, or you're doing a lot of things in math, the math parameters shift on you, because you have some very smart PhDs out there that can outdo their analysis from a math perspective, and say hey if we did this mathematically would that make a difference? Now you blend that into the principles that go with the math of cryptography, now you're trying to say well where's the clues? And I mentioned substitution, there are clues in substitution, process that allows you mathematically to address the problem without knowing cryptography per se. It's a math problem. So if you can do that and then you get into a lot of other things, what they call linear and non-linear math. Linear is a direct substitution, non-linear is a quadratic way of looking at it. What it all amounts to is it's more difficult as you're getting into the math of the cryptography. What I was trying to present, or which was presented is the different steps of looking at

this puzzle in mathematics and the basis of languages. That's if you're going to keep it as a pencil and paper system.

Ed Scheidt:

In today's genre for cryptography no one thinks of it, I mean literally no one thinks of it in the basic cryptographic education type of scenario, that's not the same. You're into the world of algorithms, which is again another step in the math process. So it evolves, it can evolve as a puzzle, it can be used in any one of the steps, it can be used as a basis of your puzzle. Which it was, you've got K1, two, three, which is steps of the puzzle, and then K4 being the final

AJ:

I think this person thought that that phrase, can you meet at the Tiger's Den at 7.30 am might have something to do with the-

Ed Scheidt:

Yeah, well the error is looking at the key which is another acronym used in cryptography to say, if there is a secret method would that method show in a key as the name implies, which opens the lock?

AJ:

Right.

Ed Scheidt:

It can be looked at in that context. Whereas the context has important facet of how you address these questions. It took a different context than the person who you're talking to.

AJ:

Another question was, did you work on any other puzzles that were outside of the CIA before you worked with Jim? Or was that the first time you were consulted?

Ed Scheidt:

That was the first time I was consulted.

AJ:

All right do you have questions for me?

Ed Scheidt:

No, I would like to read your book though when it comes out.